# Quadratic Probing

## 1. Concept

**Definition**

Quadratic Probing is an open addressing technique in which the probe sequence grows quadratically instead of linearly.

Instead of adding +1 each time, we use:

$$h'(k) = k$$

$$h(k, i) = (h'(k) + C_1 i + C_2 i^2) \bmod m$$

For this example:

$$C_1 = 1, \quad C_2 = 3$$

$$\boxed{h(k, i) = (k + i + 3i^2) \bmod 11}$$

Where:

- $k$ = key
- $i$ = probe number
- $m = 11$

## 2. Why Quadratic Probing?

Linear probing causes:

$$+1, +1, +1, +1, \ldots$$

This creates large continuous clusters (Primary Clustering).

Quadratic probing spreads probes non-linearly:

$$+1, +4, +9, +16, \dots$$

This reduces clustering.

## 3. Simulation Example

$$m = 11$$

Insert keys:

$$10, 22, 31, 4, 15, 28, 17, 88, 89$$

Initial probe:

$$h(k, 0) = k \bmod 11$$

## 4. Step-by-Step Insertion

**1) Insert 10**

$$h(10, 0) = 10$$

Insert at 10 (1 probe)

**2) Insert 22**

$$h(22, 0) = 0$$

Insert at 0 (1 probe)

**3) Insert 31**

$$h(31, 0) = 9$$

Insert at 9 (1 probe)

**4) Insert 4**

$$h(4, 0) = 4$$

Insert at 4 (1 probe)

**5) Insert 15**

$$h(15, 0) = 4 \quad \text{(collision)}$$

$$h(15, 1) = (15 + 1 + 3) = 19 \text{ mod } 11 = 8$$

Insert at 8 (2 probes)

**6) Insert 28**

$$h(28, 0) = 6$$

Insert at 6 (1 probe)

**7) Insert 17**

$$h(17, 0) = 6 \quad \text{(collision)}$$

$$h(17, 1) = 10 \quad \text{(collision)}$$

$$h(17, 2) = 9 \quad \text{(collision)}$$

$$h(17, 3) = 3$$

Insert at 3 (4 probes)

**8) Insert 88**
Since:

$$88 \text{ mod } 11 = 0$$

We simplify:

$$h(88, i) = (i + 3i^2) \text{ mod } 11$$

Compute sequentially:

$$i = 0 \rightarrow 0(collision)$$
$$i = 1 \rightarrow 4(collision)$$

$$i = 2 \to 3(collision)$$
$$i = 3 \to 8(collision)$$
$$i = 4 \to 8(collision)$$
$$i = 5 \to 3(collision)$$
$$i = 6 \to 4(collision)$$
$$i = 7 \to 0(collision)$$
$$i = 8 \to 2$$

Index 2 is empty.

Insert at 2.

Probes = 9

**9) Insert 89**

$$89 \bmod 11 = 1$$

$$h(89, 0) = 1$$

Index 1 empty.

Insert (1 probe)

# 5. Final Insertion Table

| Key | Final Index | Probes |
|---|---|---|
| 10 | 10 | 1 |
| 22 | 0 | 1 |
| 31 | 9 | 1 |
| 4 | 4 | 1 |
| 15 | 8 | 2 |
| 28 | 6 | 1 |
| 17 | 3 | 4 |
| 88 | 2 | 9 |
| 89 | 1 | 1 |

$$\boxed{\text{Total Probes} = 21}$$

$$\boxed{\alpha = \frac{9}{11} \approx 0.82}$$

## 6. Final Hash Table

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 22 | 89 | 88 | 17 | 4 | | 28 | | 15 | 31 | 10 |

## 7. Clustering Behavior

**Observation**

Quadratic probing reduces primary clustering, but keys with same initial hash follow identical probe sequences (Secondary Clustering).

## 8. Important Theoretical Points

- Quadratic probing may not visit all table slots.
- Proper choice of $m$, $C_1$, $C_2$ is important.
- Often works well when:

$$\alpha \leq 0.5$$

- Does not completely eliminate clustering.

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## Quadratic Probing = Less Clustering but More Complex Probing